Ehtimollik modellari: N-grammalar va so'z turkumlari

Tabiiy tilni qayta ishlash — 5-ma'ruza

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Prezentatsiya rejasi

Takrorlash: o'tgan darsdan ikki savol

1-savol

L4 noisy channel da $\hat{w} = \arg\max P(x|w)\,P(w)$ edi. Bu yerdagi $P(w)$ — "til modeli" — bir so'z ehtimoli nechanchi tartibli N-gram?

2-savol

P4 da `edit_distance` dinamik dasturlash (DP) bilan hisoblandi. Bugungi Viterbi ham DP — sizningcha nega ikkalasi DP?

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1-javob

Unigram (1-gram) — so'z chastotasi. Bugun uni bigram/trigramga kengaytiramiz: kontekstni hisobga olamiz.

2-javob

Ikkalasi ham masalani **kichik qism-masalalar**ga bo'lib, ularning optimal yechimidan butunni quradi — DP mohiyati.

Siz bugungi dars oxirida quyidagilarni bajara olasiz:

1. **Keltirib chiqara olasiz:** bigram ehtimolini sanash asosida (MLE) — zanjir qoidasi va Markov taxminidan.
2. **Hisoblay olasiz:** add-1 (Laplace) smoothing va perplexity ni kichik korpusda qo'lda.
3. **Qo'llay olasiz:** Markov zanjiri bilan teg ketma-ketligi ehtimolini.
4. **Keltirib chiqara olasiz:** Viterbi DP bilan eng ehtimoliy teg ketma-ketligini ($\delta(\text{VB}, t{=}2) = 0.3402$).

L2 dan: ehtimollik, $\arg\max$, log. L4 dan: dinamik dasturlash, til modeli $P(w)$. Matematikadan: ko'paytma, maksimum.

Bugungi dars rejasi:

Tsikl	**Mavzu**	**Vaqt**	**Hosil bo'luvchi natija**
1	N-gram til modeli (uni/bi/trigram)	18 daq	$P(\text{kitob}\|\text{men}) = 2/3$
2	Perplexity va smoothing	18 daq	Add-1, perplexity
3	So'z turkumlari (POS) + Markov	14 daq	$P(\text{NN}, \text{VB}) = 0.42$
4	Yashirin Markov + Viterbi	20 daq	$\delta(\text{VB}, t{=}2) = 0.3402$
	Xulosa va 5-amaliyotga ko'rsatmalar	10 daq	Ertaga nima qurishingiz

Bu dars 2-haftani ochadi

Klassik pipeline (1-hafta) dan ehtimollik modellari (2-hafta) ga o'tamiz. Bugungi Viterbi $\delta(\text{VB}, t{=}2) = 0.3402$ ertangi P5 da `assert` bo'ladi.

Bugungi dars uchun muammo: “men ___” dan keyin qaysi so’z keladi?

Ikki vazifa:

1. Avtomatik to’ldirish

Telefon klaviaturasi `"men"` dan keyin keyingi so’zni taklif qiladi. Qaysi so’z eng ehtimoliy?

2. So’z turkumini aniqlash

`"yozdi"` — bu fe’lmi yoki ot? Kontekst hal qiladi. Butun gapdagi **teglar ketma-ketligini** qanday topamiz?

Bugungi yechimlar:

1. **N-gram til modeli**: oldingi so’z(lar)ga qarab keyingisini ehtimollik bilan bashorat qiladi.
2. **HMM + Viterbi**: yashirin teglar ketma-ketligining eng ehtimoliysini topadi.

Ikkalasining yuragida bitta g’oya: **ehtimollik** va uni sanash asosida baholash.

Prezentatsiya rejasi

Intuitsiya: keyingi so'zni oldingilariga qarab bashorat qilamiz

`"men"` $\rightarrow$ `kitob? non? keldim?`

Til modeli har bir davom uchun **ehtimol** beradi va eng yuqorisini taklif qiladi.

Nechta oldingi so'z?

- **unigram**: kontekstsiz — $P(w)$
- **bigram**: 1 oldingi — $P(w_i|w_{i-1})$
- **trigram**: 2 oldingi — $P(w_i|w_{i-2}, w_{i-1})$

Ko'proq kontekst $\Rightarrow$ aniqroq bashorat, lekin ko'proq ma'lumot kerak (kamdan-kam uchraydigan N-gramlar). Bu murosani 2-tsiklda ko'ramiz.

N-gram til modeli: rasmiy ta'rif

$$P(w_i \mid w_{i-1}) = \frac{\text{count}(w_{i-1}, w_i)}{\text{count}(w_{i-1})}$$

bunda: $\text{count}(w_{i-1}, w_i)$ — w_{i-1} dan keyin w_i kelgan hollar soni; $\text{count}(w_{i-1})$ — w_{i-1} ning umumiy soni; MLE — maksimal extimollik bahosi (chastotaga asoslangan). **Gap ehtimoli (bigram):**

$$P(w_1 \dots w_n) \approx \prod_{i=1}^{n} P(w_i \mid w_{i-1})$$

Trigram: maxrajda 2 so'zli kontekst $\text{count}(w_{i-2}, w_{i-1})$.

Keltirib chiqarish: zanjir qoidasidan bigramga

1-qadam. Zanjir (ehtimollar ko'paytmasi) qoidasi — aniq:

$$P(w_1 \dots w_n) = \prod_{i=1}^{n} P(w_i \mid w_1 \dots w_{i-1})$$

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2-qadam. Muammo: $P(w_i \mid w_1 \dots w_{i-1})$ ni sanash mumkin emas — bunday uzun kontekst deyarli takrorlanmaydi (ma'lumot yetmaydi).

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2-qadam. Muammo: $P(w_i \mid w_1 \dots w_{i-1})$ ni sanash mumkin emas — bunday uzun kontekst deyarli takrorlanmaydi (ma'lumot yetmaydi). **3-qadam. Markov taxmini**: faqat oxirgi $n-1$ so'z muhim:

$$P(w_i \mid w_1 \dots w_{i-1}) \approx P(w_i \mid w_{i-1}) \quad \text{(bigram)}$$

Xulosa: aniqlikni soddalik va sanaluvchanlikka almashtiramiz. Bu — til modellashtirishning asosiy murosasi.

Qo'lda misol: bigram ehtimolini sanaymiz

S_1: `men kitob o'qidim` S_2: `men non yedim` S_3: `men kitob oldim`

Sanaymiz:

$$\text{count}(\text{men}) = 3$$
$$\text{count}(\text{men}, \text{kitob}) = 2$$

$$P(\text{kitob} \mid \text{men}) = \frac{2}{3} \approx \mathbf{0.667}$$

"men" dan keyin `kitob` (2 marta) `non` (1 marta) dan ko' proq kutiladi $\Rightarrow$ avtomatik to'ldirish `kitob` ni taklif qiladi.

Bu korpusning lug'at hajmi $|V| = 6$ (men, kitob, o'qidim, non, yedim, oldim).

Sizning vazifangiz: yana bir bigram

Savol — 90 soniya

Xuddi shu korpus (S_1, S_2, S_3) uchun:

$P(\textbf{non} \mid \textbf{men}) = ?$

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$P(\textbf{non} \mid \textbf{men}) = ?$

Javob

$\text{count}(\text{men}, \text{non}) = 1, \quad \text{count}(\text{men}) = 3$

$P(\text{non} \mid \text{men}) = \frac{1}{3} \approx \mathbf{0.333}$

Tekshiruv: $P(\text{kitob}|\text{men}) + P(\text{non}|\text{men}) = 2/3 + 1/3 = 1$ — "men" dan keyin faqat shu ikki so'z kelgan.

Kod $\leftrightarrow$ formula: bigram modeli

```python
from collections import Counter

corpus = [["men", "kitob", "o'qidim"],
          ["men", "non", "yedim"],
          ["men", "kitob", "oldim"]]

uni = Counter(w for s in corpus for w in s)
bi = Counter((s[i], s[i+1])
             for s in corpus
             for i in range(len(s) - 1))

def p_bigram(prev, w):
    return bi[(prev, w)] / uni[prev]

print(p_bigram("men", "kitob"))   # 0.667
```

Kod $\rightarrow$ formula:

Kod	Formula
`bi[(prev,w)]`	$\mathrm{count}(w_{i-1}, w_i)$
`uni[prev]`	$\mathrm{count}(w_{i-1})$
`bi/uni`	$P(w_i \mid w_{i-1})$

Ertangi P5 da `m05 Autocomplete` aynan shu sanashga tayanadi.

N-gram tuzog'i: ko'rilmagan birikma butun gapni nolga aylantiradi

Nol muammosi (zero problem)

Test gapi: `"men non o'qidim"`.

$(\text{non}, \text{o'qidim})$ korpusda **hech uchramagan** $\Rightarrow$ $P = 0$.

Bigram ko'paytmada bitta 0 $\Rightarrow$ **butun gap** ehtimoli $= 0$.

Bu jiddiy: model ko'rmagan har qanday yangi birikmani "imkonsiz" deb hisoblaydi — til esa cheksiz ijodiy.

Yechim: *smoothing* — ko'rilmagan birikmaga ham kichik ehtimol beramiz (2-tsikl).

Prezentatsiya rejasi

Intuitsiya: model qanchalik “hayron” bo’ladi?

Perplexity — model test matnini ko’rganda qanchalik “hayron” bo’lishi. Past perplexity $\Rightarrow$ yaxshi model (matnni kutgan).

Taxminan: model har qadamda **nechta** teng ehtimolli variant orasidan tanlayotgandek. Past son $\Rightarrow$ ishonchli.

> Ko’rilmagan birikmaga ham **kichik** ehtimol beramiz — ehtimollik massasining bir qismini “zaxiraga” ajratamiz. Eng oddiysi: har sanoqqa $+1$ (Laplace).

Perplexity va add-1 (Laplace) smoothing: ta'rif

$$\mathrm{PP}(W) = P(w_1 \ldots w_N)^{-\frac{1}{N}} = \exp\Big(-\tfrac{1}{N}\sum_{i=1}^{N} \ln P(w_i \mid w_{i-1})\Big)$$

bunda: N — test tokenlar soni; past PP — yaxshi model; bir tekis (uniform) $|V|$ variantli model uchun $\mathrm{PP} = |V|$.

$$P_{+1}(w_i \mid w_{i-1}) = \frac{\mathrm{count}(w_{i-1}, w_i) + 1}{\mathrm{count}(w_{i-1}) + |V|}$$

Keltirib chiqarish: nega maxrajga $|V|$ qo'shamiz?

1-qadam. Har bigram sanog'iga $+1$ qo'shamiz (ko'rilmaganlar ham 0 emas, 1 bo'ladi).

Keltirib chiqarish: nega maxrajga $|V|$ qo'shamiz?

1-qadam. Har bigram sanog'iga $+1$ qo'shamiz (ko'rilmaganlar ham 0 emas, 1 bo'ladi).
2-qadam. Ehtimollar yig'indisi 1 bo'lishi shart. w_{i-1} dan keyin **har** bir lug'at so'zi ($|V|$ ta) uchun $+1$ qo'shdik:

$$\sum_{w} (\text{count}(w_{i-1}, w) + 1) = \text{count}(w_{i-1}) + |V|$$

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$$\sum_{w} (\text{count}(w_{i-1}, w) + 1) = \text{count}(w_{i-1}) + |V|$$

3-qadam. Shuning uchun maxraj $\text{count}(w_{i-1}) + |V|$:

$$P_{+1}(w_i \mid w_{i-1}) = \frac{\text{count}(w_{i-1}, w_i) + 1}{\text{count}(w_{i-1}) + |V|}$$

Xulosa: massaning bir qismi ko'rilganlardan ko'rilmaganlarga ko'chadi — shuning uchun seen birikma ehtimoli biroz **pasayadi**.

Qo'lda misol: smoothing nol muammosini hal qiladi

$|V| = 6, \quad \text{count}(\text{non}) = 1, \quad \text{count}(\text{non}, \text{o'qidim}) = 0.$

Smoothingsiz:

$$P(\text{o'qidim} \mid \text{non}) = \frac{0}{1} = \mathbf{0}$$

$\Rightarrow$ gap ehtimoli 0, $\quad \mathrm{PP} = \infty$ (halokat).

Add-1 bilan:

$$P_{+1}(\text{o'qidim} \mid \text{non}) = \frac{0+1}{1+6} = \frac{1}{7} \approx \mathbf{0.143}$$

$\Rightarrow$ kichik, ammo **noldan katta**; PP chekli.

Smoothing modelni "yangilik"ka tayyor qiladi — ko'rilmagan birikma endi butun gapni o'ldirmaydi.

Sizning vazifangiz: add-1 ehtimolini hisoblang

Savol — 90 soniya

Xuddi shu korpus ($|V| = 6$, $\text{count}(\text{men}) = 3$, $\text{count}(\text{men}, \text{kitob}) = 2$):

$$P_{+1}(\textbf{kitob} \mid \textbf{men}) = ?$$

Smoothingsiz qiymat $2/3$ edi — add-1 dan keyin qanday o'zgaradi?

Sizning vazifangiz: add-1 ehtimolini hisoblang

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Smoothingsiz qiymat $2/3$ edi — add-1 dan keyin qanday o'zgaradi?

Javob

$$P_{+1}(\text{kitob} \mid \text{men}) = \frac{2+1}{3+6} = \frac{3}{9} = \frac{1}{3} \approx \mathbf{0.333}$$

Seen birikma ehtimoli $2/3 \to 1/3$ ga **pasaydi** — massa ko'rilmaganlarga ko'chdi. Bu smoothing narxi.

Kod ↔ formula: add-1 va perplexity

```python
import math

V = len(uni)    # lug'at hajmi = 6

def p_add1(prev, w):
    return (bi[(prev, w)] + 1) / (uni[prev] + V
        )

def perplexity(pairs):    # pairs: [(prev, w),
    ...]
    log_sum = sum(math.log(p_add1(a, b))
                  for a, b in pairs)
    return math.exp(-log_sum / len(pairs))

print(p_add1("non", "o'qidim"))    # 1/7 = 0.143
```

Kod → formula:

Kod	Formula
`+ 1`	$+1$ (Laplace)
`+ V`	$+\lvert V \rvert$
`-log_sum/N`	$-\frac{1}{N}\sum \ln P$
`exp(...)`	PP

Log fazoda yig'amiz — ko'paytma underflow ga olib keladi (4-tsikl [L]).

Perplexity tuzog'i: nimani taqqoslayapsiz?

Cheklovlar

- Perplexity faqat **bir xil lug'at** va **bir xil test to'plam**da taqqoslanadi.
- OOV (lug'atdan tashqari) so'zlar perplexity ni keskin buzadi.
- Ortiqcha smoothing taqsimotni tekislaydi — model "hammasi mumkin" deydi, foydasi kamayadi.

Past perplexity — **vosita**, maqsad emas. Yakuniy vazifadagi sifat (masalan to'ldirish to'g'riligi) muhimroq.

Yaxshiroq usullar: add-k, Kneser–Ney, backoff (kurs doirasidan tashqari).

Prezentatsiya rejasi

Intuitsiya: har so'zga turkum, har turkum oldingisiga bog'liq

So'z turkumini aniqlash (POS): har so'zga teg — ot (NN), fe'l (VB), sifat...

`men`(PR) `kitob`(NN) `o'qidim`(VB)

Markov g'oyasi: keyingi teg asosan **oldingi** tegga bog'liq. Otdan keyin ko'pincha fe'l, sifatdan keyin ot...

Bir xil so'z turli teg olishi mumkin: kontekst hal qiladi. Markov zanjiri teglar orasidagi **o'tish**larni modellaydi.

Markov zanjiri: rasmiy ta'rif

$$P(s_1 s_2 \dots s_T) = \pi(s_1) \prod_{t=2}^{T} A(s_t \mid s_{t-1})$$

bunda: s_t — t-holat (teg); $\pi(s_1)$ — boshlang'ich ehtimol; $A(s_t \mid s_{t-1})$ — o'tish (transition) ehtimoli; 1-tartib — keyingi holat faqat **bitta** oldingisiga bog'liq. **Bizning o'tish matritsasi (NN, VB):**

	$\to$ NN	$\to$ VB
NN	0.4	0.6
VB	0.3	0.7

$\pi(\text{NN}) = 0.7, \ \pi(\text{VB}) = 0.3$

Keltirib chiqarish: to'liq qo'shma ehtimoldan Markovgacha

1-qadam. Teglar ketma-ketligining to'liq ehtimoli (zanjir qoidasi):

$$P(s_1 \ldots s_T) = \pi(s_1) \prod_{t=2}^{T} P(s_t \mid s_1 \ldots s_{t-1})$$

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2-qadam. 1-tartibli Markov taxmini — faqat oldingi holat muhim:

$$P(s_t \mid s_1 \ldots s_{t-1}) \approx P(s_t \mid s_{t-1}) = A(s_t \mid s_{t-1})$$

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2-qadam. 1-tartibli Markov taxmini — faqat oldingi holat muhim:

$$P(s_t \mid s_1 \dots s_{t-1}) \approx P(s_t \mid s_{t-1}) = A(s_t \mid s_{t-1})$$

3-qadam. Demak:

$$P(s_1 \dots s_T) = \pi(s_1) \prod_{t=2}^{T} A(s_t \mid s_{t-1})$$

Bu — N-gram bilan **bir xil g'oya**, faqat so'zlar o'rniga teglar ustida.

Qo'lda misol: teg ketma-ketligi ehtimoli

$$\pi(\text{NN}) = 0.7, \quad A(\text{VB} \mid \text{NN}) = 0.6.$$

Hisoblaymiz:

$$\begin{aligned} P(\text{NN}, \text{VB}) &= \pi(\text{NN}) \cdot A(\text{VB} \mid \text{NN}) \\ &= 0.7 \times 0.6 = \mathbf{0.42} \end{aligned}$$

Bu faqat **teglar** ehtimoli (so'zlarsiz). Keyingi tsiklda so'zlar (kuzatuvlar) qo'shilib, HMM hosil bo'ladi.

Sizning vazifangiz: teskari ketma-ketlik

Savol — 60 soniya

$\pi(\mathrm{VB}) = 0.3, \quad A(\mathrm{NN} \mid \mathrm{VB}) = 0.3.$

$P(\mathbf{VB}, \mathbf{NN}) = ?$

Sizning vazifangiz: teskari ketma-ketlik

Savol — 60 soniya

$\pi(\text{VB}) = 0.3, \quad A(\text{NN} \mid \text{VB}) = 0.3.$

$P(\textbf{VB}, \textbf{NN}) = ?$

Javob

$P(\text{VB}, \text{NN}) = \pi(\text{VB}) \cdot A(\text{NN} \mid \text{VB}) = 0.3 \times 0.3 = \mathbf{0.09}$

$0.09 < 0.42$ — bu korpusda gap **ot bilan boshlanib fe'l bilan davom etishi** ancha ehtimoliyroq (NN$\to$VB).

```python
pi = {"NN": 0.7, "VB": 0.3}
A = {("NN", "NN"): 0.4, ("NN", "VB"): 0.6,
     ("VB", "NN"): 0.3, ("VB", "VB"): 0.7}

def seq_prob(tags):
    p = pi[tags[0]]
    for i in range(1, len(tags)):
        p *= A[(tags[i-1], tags[i])]
    return p

print(seq_prob(["NN", "VB"]))   # 0.42
print(seq_prob(["VB", "NN"]))   # 0.09
```

Kod → formula:

Kod	Formula
`pi[tags[0]]`	$\pi(s_1)$
`A[(prev,cur)]`	$A(s_t \mid s_{t-1})$
`p *= ...`	$\prod$

A — o'tish matritsasi lug'at sifatida. P5 da `m05b POSTagger` shuni kengaytiradi.

Markov tuzog'i: uzoq bog'liqlik ko'rinmaydi

1-tartib cheklovi

Keyingi holat faqat **bitta** oldingisiga bog'liq. Lekin:

"agar yomg'ir yog'sa... biz uyda qolamiz"

"qolamiz" "agar" ga bog'liq — ammo ular orasida ko'p so'z bor. 1-tartib Markov buni **ko'rmaydi**.

Yuqori tartib (2-, 3-) biroz yordam beradi, lekin ma'lumot talabini oshiradi.

To'liq yechim: uzoq bog'liqlikni ushlaydigan modellar — LSTM (2-hafta oxiri), Transformer (3-hafta).

Prezentatsiya rejasi

Intuitsiya: teglar yashirin, so'zlar ko'rinadi

Biz **so'zlar**ni ko'ramiz (`nlp`, `yozdi`), lekin **teglar** (NN, VB) yashirin. HMM:

- teglar Markov zanjiri bo'ylab o'tadi (A),
- har teg so'zni "chiqaradi" (emission, B).

Vazifa: so'zlar berilgan, eng ehtimoliy teg ketma-ketligini topish.

Barcha teg yo'llarini sinash eksponensial (2^T). **Viterbi** — DP bilan eng yaxshi yo'lni $O(T \cdot S^2)$ da topadi.

$$\delta_t(j) = \left[\max_i \ \delta_{t-1}(i)\, A(j \mid i)\right] B(o_t \mid j)$$

bunda: HMM $= (\pi, A, B)$: π — boshlang'ich, A — o'tish, $B(o_t \mid j)$ — j holat o_t so'zni chiqarish (emission) ehtimoli; $\delta_t(j)$ — t da j da tugaydigan eng ehtimoliy yo'l qiymati. **Asos va yo'l:**
$\delta_1(j) = \pi(j)\, B(o_1 \mid j)$; backpointer $\psi_t(j) = \arg\max_i \delta_{t-1}(i)A(j \mid i)$ orqali teskari yuramiz.

Keltirib chiqarish: nega Viterbi DP ishlaydi?

1-qadam. Maqsad: $\arg\max_{s_1 \ldots s_T} P(s_1 \ldots s_T, o_1 \ldots o_T)$. Barcha yo'llar — S^T ta (juda ko'p).

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1-qadam. Maqsad: $\arg\max_{s_1 \dots s_T} P(s_1 \dots s_T, o_1 \dots o_T)$. Barcha yo'llar — S^T ta (juda ko'p).
2-qadam. Kuzatuv: t da j da tugaydigan eng yaxshi yo'l, faqat $t-1$ dagi eng yaxshi yo'llarga tayanadi (Markov xossasi).

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1-qadam. Maqsad: $\arg\max_{s_1\ldots s_T} P(s_1\ldots s_T, o_1\ldots o_T)$. Barcha yo'llar — S^T ta (juda ko'p). **2-qadam.** Kuzatuv: t da j da tugaydigan eng yaxshi yo'l, faqat $t-1$ dagi eng yaxshi yo'llarga tayanadi (Markov xossasi). **3-qadam.** Shuning uchun har qadamda har holat uchun eng yaxshi kirishni saqlaymiz:

$$\delta_t(j) = \left[\max_i \delta_{t-1}(i) A(j \mid i)\right] B(o_t \mid j)$$

Xulosa: eksponensial qidiruv $O(TS^2)$ ga tushadi — edit_distance (L4) bilan bir xil DP mantiqi.

Qo'lda hisob: Viterbi — $\delta(\text{VB}, t{=}2) = 0.3402$

$\pi(\text{NN}){=}0.7$, $\pi(\text{VB}){=}0.3$; $B(\text{nlp}|\text{NN}){=}0.9$, $B(\text{nlp}|\text{VB}){=}0.1$, $B(\text{yozdi}|\text{NN}){=}0.1$, $B(\text{yozdi}|\text{VB}){=}0.9$; $A(\text{VB}|\text{NN}){=}0.6$, $A(\text{NN}|\text{VB}){=}0.3$, $A(\text{NN}|\text{NN}){=}0.4$, $A(\text{VB}|\text{VB}){=}0.7$

$t{=}1$ **(nlp):**

$$\delta_1(\text{NN}) = 0.7 \cdot 0.9 = 0.63$$
$$\delta_1(\text{VB}) = 0.3 \cdot 0.1 = 0.03$$

	$t{=}1$ (nlp)	$t{=}2$ (yozdi)
NN	0.63	0.0252
VB	0.03	**0.3402***

$t{=}2$ **(yozdi), VB:**

$$\delta_2(\text{VB}) = \max(0.63{\cdot}0.6,\ 0.03{\cdot}0.7) \cdot 0.9$$
$$= 0.378 \cdot 0.9 = \mathbf{0.3402}$$

Backtrace: $\delta_2(\text{VB})$ NN dan keldi (0.378) $\Rightarrow$ ketma-ketlik **[NN, VB]**.

Ertangi P5 birinchi assert:

```
abs(delta_VB_t2 - 0.3402) < 1e-4
```

 # Ma'ruza L5 [I4]-slayd

Sizning vazifangiz: $\delta_2(\mathrm{NN})$ ni hisoblang

Savol — 2 daqiqa

Xuddi shu HMM uchun $t{=}2$ (yozdi) da **NN** qiymati:

$$\delta_2(\mathbf{NN}) = \max\Big(\delta_1(\mathbf{NN})A(\mathbf{NN}|\mathbf{NN}),\ \delta_1(\mathbf{VB})A(\mathbf{NN}|\mathbf{VB})\Big)\cdot B(\mathbf{yozdi}|\mathbf{NN}) = ?$$

Sizning vazifangiz: $\delta_2(\mathrm{NN})$ ni hisoblang

Savol — 2 daqiqa

Xuddi shu HMM uchun t=2 (yozdi) da **NN** qiymati:

$$\delta_2(\mathbf{NN}) = \max\Big(\delta_1(\mathbf{NN})A(\mathbf{NN}|\mathbf{NN}),\ \delta_1(\mathbf{VB})A(\mathbf{NN}|\mathbf{VB})\Big) \cdot B(\mathbf{yozdi}|\mathbf{NN}) = ?$$

Javob

$\max(0.63{\cdot}0.4,\ 0.03{\cdot}0.3) \cdot 0.1 = \max(0.252, 0.009) \cdot 0.1 = 0.252 \cdot 0.1 = \mathbf{0.0252}$

t=2 da $\delta(\mathrm{VB}) = 0.3402 > \delta(\mathrm{NN}) = 0.0252 \Rightarrow$ oxirgi teg **VB**; backtrace **[NN, VB]**.

Kod $\leftrightarrow$ formula: Viterbi

```python
def viterbi(obs, states, pi, A, B):
    delta = [{s: pi[s]*B[(obs[0], s)] for s in states}]
    back = [{}]
    for t in range(1, len(obs)):
        d, bk = {}, {}
        for j in states:
            i = max(states,
                    key=lambda i: delta[t-1][i]*A[(i, j)])
            d[j] = delta[t-1][i]*A[(i, j)]*B[(obs[t], j)]
            bk[j] = i
        delta.append(d); back.append(bk)
    last = max(states, key=lambda s: delta[-1][s])
    path = [last]
    for t in range(len(obs)-1, 0, -1):
        last = back[t][last]; path.insert(0, last)
    return path, delta
```

Kod $\to$ formula:

Kod	Formula
`pi[s]*B`	$\delta_1(j)$
`max(...A)`	$\max_i \delta A$
`*B[(o,j)]`	$\cdot B(o_t \vert j)$
`bk[j]=i`	$\psi_t(j)$

P5 da `m05b POSTagger` aynan shu Viterbi ni amalga oshiradi.

Viterbi tuzog'i: ehtimollar ko'paytmasi nolga intiladi

Underflow

Uzun gapda ko'p kichik ehtimol ko'paytiriladi $(0.9 \cdot 0.1 \cdot 0.6 \cdots) \Rightarrow$ son shunchalik kichrayadiki, kompyuter uni 0 deb yumaloqlaydi.

Yechim: log fazo

Ko'paytma o'rniga **logarifmlar yig'indisi**: $\log(ab) = \log a + \log b$. Maksimum tartibi saqlanadi.

Yana bir nuqta: Viterbi eng ehtimoliy **yo'l**ni beradi — har bir holatning alohida marginal ehtimoli emas.

Prezentatsiya rejasi

O'zbek tilida: SOV tartibi va agglutinatsiya N-gramni qiynaydi

1. SOV tartibi (kesim oxirida):

"men kitobni do'stimga berdim"

Kesim (`berdim`) **oxirida**. Bigram "men ___" keyingi so'zni taxmin qiladi, lekin gap ma'nosini belgilovchi fe'l uzoqda ⇒ uzoq bog'liqlik (L3 tuzog'i).

2. Agglutinatsiya:

`kitob`, `kitobni`, `kitobga`, `kitoblar` — har biri **alohida** token.

Oqibat

N-gram lug'ati **portlaydi** ($|V|\uparrow$); ko'pchilik N-gramlar 0 yoki 1 marta uchraydi ⇒ siyrak, smoothing ayniqsa muhim.

Yechim: stemming (m01) N-gramdan oldin lug'atni kichraytiradi.

Sintez: kontekst qancha bo'lsa, ma'lumot shuncha kerak

Model	**Kontekst**	**Murosa**
Unigram	yo'q	juda kam ma'lumot; kontekst yo'q
Bigram	1 oldingi so'z	muvozanat (kurs tanlovi)
Trigram	2 oldingi so'z	aniqroq, ammo siyrak
HMM (Viterbi)	yashirin teglar	ketma-ketlik teglash

Asosiy g'oya

Hammasining yuragida: **ehtimollikni sanash** + **Markov taxmini** + (kerakli joyda) **smoothing**. Viterbi — shu g'oyalarni ketma-ketlik teglashga DP bilan qo'llaydi.

Seminal manba: Rabiner (1989)

Rabiner, L.R. (1989). **A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition.** *Proceedings of the IEEE*, 77(2), 257–286.

Ahamiyati:

- HMM uchun "oltin standart" qo'llanma — 30 000+ iqtibos.
- Uch masala: baholash (Forward), dekodlash (Viterbi), o'rganish (Baum–Welch).
- Nutq, POS, bioinformatika — hammasi shu doirada.

Muhokama savoli (P5 gacha o'ylang)

HMM teglash uchun **teglangan korpus** kerak. O'zbek tili uchun bunday korpus kam. Bu cheklovni qanday yengish mumkin? (Maslahat: 4-hafta, BERT.)

Bugungi maqsadlar bajarildi

1. ✓ **Keltirib chiqardik:** bigram MLE — $P(\text{kitob}|\text{men}) = 2/3$ (zanjir qoidasi + Markov taxmini).
2. ✓ **Hisobladik:** add-1 smoothing ($P_{+1}(\text{o'qidim}|\text{non}) = 1/7$) va perplexity.
3. ✓ **Qo'lladik:** Markov zanjiri — $P(\text{NN}, \text{VB}) = 0.42$.
4. ✓ **Keltirib chiqardik:** Viterbi DP — $\delta(\text{VB}, t{=}2) = 0.3402$, ketma-ketlik [NN, VB].

Ertaga P5: avtomatik to'ldirish va POS teglash quriladi

N-gram model + smoothing → `m05` **Autocomplete**

HMM + Viterbi → `m05b` **POSTagger**

P5 da qurasiz:

- `m05 Autocomplete`: bigram + add-1, `perplexity`
- `m05b POSTagger`: HMM + `viterbi`
- Eng ehtimoliy davom va teg ketma-ketligi

Birinchi assert (bu slayd [I4]):

```
assert abs(delta_VB_t2
       - 0.3402) < 1e-4
```

 # Ma'ruza L5 [I4]-slayd bilan
 # solishtiring; yo'l == [NN, VB]

Adabiyotlar

1. Rabiner, L.R. (1989). *A Tutorial on Hidden Markov Models...* Proc. IEEE 77(2):257–286.
2. Jurafsky, D., Martin, J.H. (2023). *Speech and Language Processing*, 3-nashr. 3-bob: N-gram LM; 8-bob: POS, HMM.
3. Shannon, C.E. (1948). *A Mathematical Theory of Communication.* (Til modellashtirish va entropiya ildizi.)
4. Chen, S., Goodman, J. (1999). *An Empirical Study of Smoothing Techniques for Language Modeling.* Computer Speech & Language.
5. `nltk.lm` hujjatlari — N-gram til modellari.

Ilova S1: log fazoda Viterbi (underflow yechimi)

Ko'paytma o'rniga logarifmlar yig'indisi:

$$\log \delta_t(j) = \max_i \left[\log \delta_{t-1}(i) + \log A(j \mid i) \right] + \log B(o_t \mid j)$$

bunda: max va arg max tartibi log monoton bo'lgani uchun saqlanadi.

Misol (t=2, VB): $\log \delta_2(\mathrm{VB}) = \max(\log 0.63 + \log 0.6,\ \log 0.03 + \log 0.7) + \log 0.9$. Bu $\ln 0.3402 \approx -1.078$ ga teng — xuddi shu natija, lekin underflowsiz.

Ilova S2: perplexity va entropiya bog'liqligi

Perplexity — model kross-entropiyasining eksponentasi:

$$\mathrm{PP}(W) = 2^{H(W)}, \qquad H(W) = -\frac{1}{N}\sum_{i} \log_2 P(w_i \mid w_{i-1})$$

bunda: $H(W)$ — o'rtacha kross-entropiya (bit/token).

Intuitsiya: agar model har qadamda k ta teng ehtimolli variant orasidan "tanlayotgan" bo'lsa, $\mathrm{PP} = k$. Shuning uchun perplexity ni "o'rtacha tarmoqlanish koeffitsiyenti" deb o'qish mumkin.